

Data and Code Availability in Sports Science: A Registered Report

Registered Report – Stage 1

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Abstract

Sharing scientific raw data and analysis code facilitates cumulative scientific work and enables assessments of reproducibility and robustness. In sports science, preliminary evidence suggests that data and code availability may be low, but previous studies have been based on small cross-sectional samples. To provide a larger audit of the literature, we will document data and code sharing in all original articles published in Q1 sport science journals over the past ten years. We will calculate the percentages of reported as well as actual public availability of data and code and investigate whether general time trends exist. Data and code sharing policies of the analyzed journals will be surveyed to examine associations between policy changes and sharing rates. We provide a resource with all articles in the sample that have shared data and/or code for reuse. The findings of this study can be used to inform future data and code sharing policies in sport science.

Keywords: research transparency; open data; open code; research materials

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1 Introduction

In light of the reproducibility crisis in psychology and other scientific disciplines and of evidence suggesting that reproducibility may be also low in sports sciences, access to the raw study data is becoming increasingly important to establish robust evidence. Thereby, sharing research data and analysis codes from empirical studies can be beneficial for several reasons. Data and code sharing enables assessments of reproducibility (whether the same analysis on the shared data yield the same results) and analytical robustness (whether reasonable deviations in analytical decisions yield similar results) (e.g., Brodeur et al., 2026). Shared data and code also allow reuse by other researchers, for example, to test alternative theories using the same dataset, combine results across studies (e.g., with individual participant meta-analyses), inform future analyses (e.g., as input parameters for power analyses or priors for Bayesian analyses), or adapt computational workflows to related research problems. From a normative perspective, sharing data and code might also be the right thing to do to advance science, particularly when the research has been publicly funded (Arzberger et al., 2006).

Nevertheless, sharing data may also be problematic in some cases. Privacy concerns, particularly in studies of small or vulnerable populations, may prohibit the sharing of data because of the risk of identifying individuals. Proprietary datasets may not be available for sharing because of commercial interests. However, methods for de-identification and simulation of pseudo-datasets have been developed (Naughton et al., 2023; Warmenhoven et al., 2025). With these methods, that arguably require additional effort, data for reuse and reproducibility analyses can be shared most of the times. For analysis code, privacy concerns usually do not apply because analysis code typically does not contain sensitive information about research participants. However, sharing analysis code requires the analysis to be conducted in a programming language, or a code-generating program, which may not always be the case. At the level of the researcher, preparing data and code for sharing may require additional work and training, which some may perceive as unnecessary if it is not incentivized.

Over the past two decades, some disciplines like psychology or ecology have seen a moderate increase in data and code sharing (Hardwicke et al., 2024; Tedersoo et al., 2021). In contrast, a meta-analysis of studies investigating data sharing practices in medicine found that, between 2014 and 2020, only reported, but not actual data availability had increased (Hamilton et al., 2023). In addition, data availability statements indicating that “data is available upon reasonable request” often mean that data will not be shared on request in practice (Gabelica et al., 2022; Hussey, 2025; Krawczyk & Reuben, 2012; Tedersoo et al., 2021). Data sharing alone does not guarantee that data can be reused (Towse et al., 2021), since it should be shared with appropriate meta-data and licenses (see the FAIR principles, Wilkinson et al., 2016).

Policies shape data and code sharing practices. Case studies of journals enforcing mandatory data sharing show increases in public data availability (Hardwicke et al., 2018; Laurinavichyute et al., 2022). Some fields have gone further: For example, all major economics journals require researchers to submit “replication packages” with data and analysis code along their articles, and these usually undergo in-house or external reproducibility checks (Brodeur et al., 2024). However, outside selected fields or selected journals, data and code sharing is generally encouraged rather than mandated (Resnik et al., 2019).

In sport science, preliminary evidence suggests that data and code availability is low. Twomey et al. (2021) found that, among 300 randomly sampled articles published in 2019 in three leading sport science journals, only two articles shared data (0.7%). Borg et al. (2020) reported a data sharing rate of 1.8% among 283 original articles randomly sampled from Q1 sport science journals in 2019. Schulz et al. (2022) found that none of the 163 clinical trials in sports medicine and orthopedics published in 2019 and 2020 shared their data. Among 195 studies in sports

medicine published between 2014 and 2018, only 2.6% shared at least some data (Evans et al., 2019). Code availability in sport science has been reported at 0% (Borg et al., 2020; Evans et al., 2019).

Previous analyses of data and code availability in sports science have been cross-sectional and limited to small random samples (≤ 300 articles). Moreover, publication practices may have changed over the last decade, as a number of articles in the field have pointed out the benefits of sharing data and/or code (Abt et al., 2022; Borg et al., 2020; Bullock et al., 2022, 2024; Halperin et al., 2018; Memmert, 2025; Sainani & Chamari, 2022; Tenan & Alejo, 2024). At present, a systematic analysis of data and code sharing practices across the field of sports science is missing. To inform decisions on future sharing policies, a description of the status quo, current trends, and indirect evidence of the effectiveness of past and current policies is needed.

The aim of this study is to descriptively present current data and code sharing practices across the full population of original articles published in Q1 sport science journals over the past ten years. We will investigate whether general time trends exist, and will document data and code sharing policies in sports science journals to exploratively analyze their association with sharing practices. Our findings will provide an assessment of the current state of data and code availability in sports science that can inform future discussions about sharing policies.

2 Methods

2.1 Article Search

All articles published in Q1 sports science journals over the past 10 years will be analyzed. We define eligible journals as those that were listed as a Q1 journal in the field “Sports Science” in the SCImago Journal Rank for at least three years between 2017 and 2025 (SCImago, 2025). Journals that only publish review articles or qualitative research will be excluded. Using Web of Science, we will search for all original research articles published in the included journals between 2017 and 2026 (see Supplementary Materials S1 for the exact search term). We focus on original articles because they are the primary source of scientific raw data. Review articles or editorials typically use aggregated data or no data at all, and thus will be excluded from the search. For all included articles, the dates of acceptance and data base indexing (used as a proxy for online publication date) are retrieved using the Crossref API via the rcrossref package for R (Chamberlain et al., 2026).

2.2 Sample Size and Sample Size Justification

The final sample is expected to consist of around 35,000 articles. This estimate assumes that 25 journals will meet the search criteria, each publishing on average 175 articles per year, of which 80% are original research articles. We do not perform a power- or precision-based sample size estimation because our sample constitutes the full population of articles published in Q1 sports science journals over the past 10 years.

2.3 Data Extraction

The full text retrieval and data extraction procedures were piloted using an independent random sample of 200 articles published in journals that were not listed as Q1 in sports science in the SCImago ranking but otherwise met the inclusion criteria. The pilot informed our estimates of the time required for article retrieval and the error rates of extraction procedures. A detailed summary of the pilot is presented in Supplementary Materials S2.

The full-text manuscripts of all eligible studies will be downloaded using the *Zotero* bulk import function, or by manually downloading the articles from the publishers’ websites. Based on the

pilot data, the bulk import is expected to succeed in approximately 25% of cases. Assuming an average time of 16.5 seconds required per article for manual retrieval of the PDF, as observed in the pilot, manually retrieving the remaining articles would require around 120 working hours. Using *Grobid* via the *metacheck* R package (DeBruine et al., 2026; *GROBID*, 2026), we will parse the PDF files to extract the article texts. An automated algorithm based on the *Grobid* section detection and regular expression search will be used to flag articles that may contain statements of data/code availability and data/code provision (see Supplementary Materials S3). All articles that are flagged by the automated approach will be manually reviewed to extract the items of interest, minimizing the rate of false positives. During the piloting, the false negative rate of the automated approach was 0%.

For each article, we will determine whether a data availability statement exists (yes/no) and classify the content of the statement (data included in repository, data included in manuscript, data available upon (reasonable) request, data cannot be made available). We document whether the raw data is made available for download. Raw data is defined as the individual participant data that are subjected to the statistical analyses of an article. The same extraction methodology will be applied for code availability and code provision.

Among the subset of articles that share data, we document whether the FAIR principles (findable, accessible, interoperable, reusable) (Wilkinson et al., 2016) are followed. We will apply a simple and liberal operationalization of the FAIR principles to assess compliance. Specifically, we will define “findable” as the presence of a persistent and unique digital identifier (e.g., DOI) and associated meta-data (e.g., authors, year, title); “accessible” as data being accessible via an open protocol (e.g., HTTP) or a documented authorization procedure; “interoperable” as data being saved in data formats that are typical for the research field and do not require specialized proprietary software to be read; and “reusable” as being provided with a license that documents the terms and conditions for reuse.

To contextualize code sharing practices, we will also document the (statistical) analysis software used in all included articles. The software used will be identified by regular expression matching against mentions of common analysis software packages in the manuscript text and references (see Supplementary Materials S3). Articles that are flagged as using the software *R* will be manually checked because regular expression matching proved difficult for these cases during the pilot. Based on this procedure, the pilot data indicate that the error rates of the automated approach are approximately 1-2% for false positives and false negatives, respectively. Combined, these error rates are lower than the false negative rate of manual extraction in the pilot (13%), and we deem them sufficient for the present analysis.

2.4 Journal Policies

All journals included in the sample will be contacted and asked for the current and past editorial guidelines regarding data and code sharing. In addition, we will review current, as well as past (via the Internet Archive, <https://web.archive.org/>) author instructions for all journals. Based on previous work (Hamilton et al., 2020) and the TOP guidelines (Nosek et al., 2015), we will classify data and code sharing policies into four levels: Level 0 (no policy), Level 1 (sharing encouraged and availability statement mandatory), Level 2 (sharing mandatory), Level 3 (FAIR sharing mandatory and/or in-house reproducibility checks).

2.5 Statistical Analysis

The primary outcome is the percentage of studies in sport science that make raw data and/or analysis code available. We will descriptively estimate this proportion by aggregating over the full ten-year sampling period without weighting.

As secondary outcomes, we will descriptively report the percentage of studies that used data availability statements, the reported data availability, as well as the percentage of studies that shared FAIR-aligned data among those that share data. Descriptive results of the analysis software usage across all articles will be presented to contextualize the findings regarding code sharing.

To investigate whether data and code sharing have changed over time, we will fit two logistic regression models of the form $\text{shared} \sim \text{time} + (\text{time} | \text{journal})$, where shared is the outcome indicating whether code/data has been shared, time is the grand-mean centered continuous indicator of time of acceptance, and time|journal is a random effect on the journal level, that allows baseline sharing rates and time trends to vary by journal. We will interpret the fixed main effect of time (odds-ratio) and its 95%-confidence interval, with an 95% confidence interval not including an odds-ratio of 1 as an indicator of the existence of a time effect.

Current journal policies will be presented descriptively. In an exploratory analysis, we will investigate the associations between changes in data or code sharing policies and the presence of data/code availability statements and shared data/code. The percentage of articles including availability statements and data/code sharing will be visually presented as a 100-day moving average for selected journals, for which the policy history can be fully reconstructed, with markers indicating changes in guidelines. This approach allows us to contrast associations between policies and data/code sharing across journals and within journals over time.

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